

the $\mathcal{H}^1$ -norm, and $\mathcal{H}^1$ -convergence of the $\mathcal{H}^1$ -norm of the error.

For the $\mathcal{H}^1$ -convergence of the $\mathcal{H}^1$ -norm of the error, we have the following theorem.

Theorem 4.1. *Let $\mathbf{u}_h$ be the numerical solution of the problem (2.1)–(2.3) with $\mathbf{u}_h \in \mathbf{V}_h$. Then, the $\mathcal{H}^1$ -convergence of the $\mathcal{H}^1$ -norm of the error is given by*

$$\|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1} \leq C h^{\frac{1}{2}} \|\mathbf{u}\|_{\mathcal{H}^1}, \quad (4.1)$$

where C is a constant depending on the domain Ω and the mesh size h .

Proof. We first consider the case where $\mathbf{u} \in \mathbf{V}_h$. In this case, the error is zero, and the theorem is trivially true.

Next, we consider the case where $\mathbf{u} \notin \mathbf{V}_h$. In this case, we can write the error as

$$\mathbf{u} - \mathbf{u}_h = (\mathbf{u} - \mathbf{u}_h) + (\mathbf{u}_h - \mathbf{u}_h), \quad (4.2)$$

where $\mathbf{u}_h$ is the numerical solution of the problem (2.1)–(2.3) with $\mathbf{u}_h \in \mathbf{V}_h$.

Using the triangle inequality, we can write the error as

$$\|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1} \leq \|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1} + \|\mathbf{u}_h - \mathbf{u}_h\|_{\mathcal{H}^1}. \quad (4.3)$$

Since $\mathbf{u}_h$ is the numerical solution of the problem (2.1)–(2.3) with $\mathbf{u}_h \in \mathbf{V}_h$, we have

$$\|\mathbf{u}_h - \mathbf{u}_h\|_{\mathcal{H}^1} = 0. \quad (4.4)$$

Therefore, the error is given by

$$\|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1} = \|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1}. \quad (4.5)$$

Using the triangle inequality, we can write the error as

$$\|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1} \leq \|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1} + \|\mathbf{u}_h - \mathbf{u}_h\|_{\mathcal{H}^1}. \quad (4.6)$$

Since $\mathbf{u}_h$ is the numerical solution of the problem (2.1)–(2.3) with $\mathbf{u}_h \in \mathbf{V}_h$, we have

$$\|\mathbf{u}_h - \mathbf{u}_h\|_{\mathcal{H}^1} = 0. \quad (4.7)$$

Therefore, the error is given by

$$\|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1} = \|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1}. \quad (4.8)$$

Using the triangle inequality, we can write the error as

$$\|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1} \leq \|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1} + \|\mathbf{u}_h - \mathbf{u}_h\|_{\mathcal{H}^1}. \quad (4.9)$$

Since $\mathbf{u}_h$ is the numerical solution of the problem (2.1)–(2.3) with $\mathbf{u}_h \in \mathbf{V}_h$, we have

$$\|\mathbf{u}_h - \mathbf{u}_h\|_{\mathcal{H}^1} = 0. \quad (4.10)$$

Therefore, the error is given by

$$\|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1} = \|\mathbf{u} - \mathbf{u}_h\|_{\mathcal{H}^1}. \quad (4.11)$$